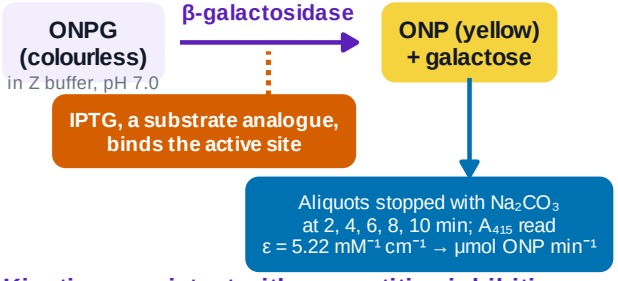
Stopped-reaction spectrophotometry of β-galactosidase: substrate saturation and kinetic behaviour consistent with competitive inhibition by IPTG

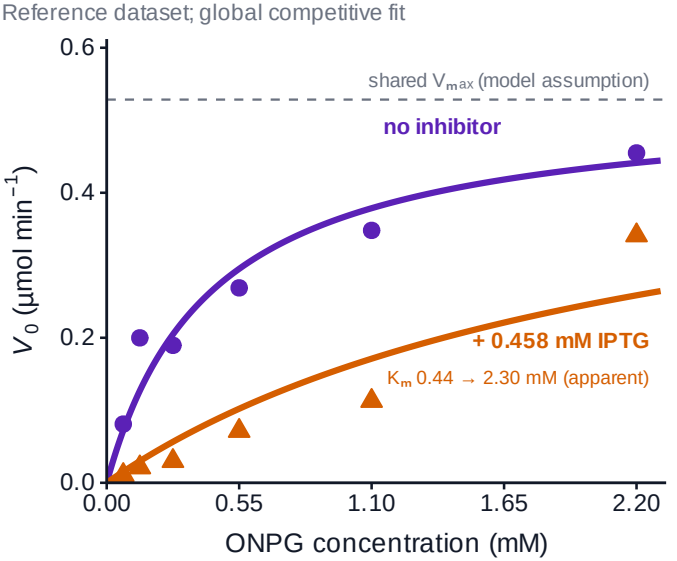


Kinetics consistent with competitive inhibition

V_0 approximately ∝ enzyme (R^2 = 0.97)

K_m = 0.31 mM, V_{max} = 0.48 µmol min⁻¹

+ 0.458 mM IPTG: apparent K_m ×5.6

K_i = 0.10–0.22 mM across rule-based estimators

(0.50 mM under the crudest; purified enzyme 0.08 mM)